

Blended RAG-Enhanced LLM Multi-Agent Framework for Anomaly Detection and Alert Management

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Abstract—The increasing complexity of next-generation power information systems poses significant challenges for anomaly detection and alarm handling. Traditional rule-based approaches often suffer from high false alarms, low recall, and limited scalability, while large language models (LLMs), despite their strong reasoning abilities, are prone to hallucinations and factual inconsistencies. To address these issues, this paper proposes a Blended RAG-Enhanced LLM Multi-Agent System for intelligent operation and maintenance. By integrating a hybrid retrieval mechanism that combines sparse and dense retrieval with a Best Fields strategy, the system improves domain knowledge recall and factual consistency. Within a multi-agent framework based on the ReAct paradigm, RAG is embedded into task decomposition, response planning, and safety validation agents, forming a closed-loop workflow of detection - decision - verification. The proposed approach enhances automation, mitigates hallucination risks, and ensures safe and reliable execution of incident handling in power O&M scenarios.

Keywords—Retrieval-Augmented Generation, Large Language Model, Multi-Agent System, Anomaly Detection

I. INTRODUCTION

With the rapid development of next-generation power systems, business operations have become increasingly complex, extending to the network edge and significantly expanding the security boundary and attack surface of information and communication infrastructures. This expansion has introduced greater vulnerability to data, networks, devices, and business processes. Consequently, anomaly analysis and incident response have become critical for ensuring the secure and stable operation of power information systems.

Traditional operation and maintenance (O&M) approaches rely heavily on a combination of rule-based engines and

operator expertise, such as fixed thresholds, pattern-matching rules, and alarm correlation strategies, to detect and classify system logs, performance indicators, and events[1]. However, due to the static and deterministic nature of these rules, such methods exhibit clear limitations when addressing dynamic, multi-system, and cross-link fault propagation patterns, often resulting in high false alarm and missed detection rates. While manual intervention allows operators to flexibly handle unexpected situations, it suffers from poor knowledge accumulation, inconsistent handling quality, and limited scalability in high-concurrency scenarios[2].

In recent years, the rapid advancement of Large Language Models (LLMs) has created new opportunities for intelligent O&M[3]. Leveraging their strong natural language understanding and generation capabilities, LLMs can flexibly analyze event descriptions, retrieve historical knowledge, and generate response plans. However, relying solely on intrinsic reasoning makes them vulnerable to hallucinations and factual inconsistencies — i.e., generating seemingly reasonable but factually incorrect instructions in the absence of real supporting data[4]. In safety-critical domains such as power operations, such risks are unacceptable. To mitigate hallucinations, Retrieval-Augmented Generation (RAG) has been introduced, which enhances reasoning by retrieving highly relevant external knowledge as contextual input[5]. Nevertheless, conventional RAG approaches mainly rely on dense retrieval, which, although strong in semantic similarity, often fails to recall domain-specific long-tail terminology or handle structured operational data, leading to mismatches between retrieval results and real operational needs[6].

To address these limitations, this paper proposes a Blended RAG-enhanced LLM multi-agent system for business anomaly detection and automated alert management. The proposed

system retains the semantic matching capabilities of dense retrieval while introducing domain-adapted sparse encoders to accurately match power-industry-specific terminology[7]. Furthermore, it incorporates a Best Fields strategy to prioritize critical fields within structured operational data[8]. Together, these mechanisms significantly improve the accuracy and adaptability of domain knowledge retrieval. Compared with conventional dense-only RAG methods, the Blended RAG approach achieves superior performance in domain-specific terminology recall, structured knowledge utilization, and factual consistency, thereby providing LLM reasoning with more precise and verifiable factual support. This effectively reduces hallucination risks and enhances both the executability and safety of generated response plans[9].

From an engineering perspective, the system integrates the Blended RAG method with a multi-agent decision-making engine, enabling a fully automated and closed-loop workflow encompassing task decomposition, plan generation, and safety validation. Additionally, industry safety and compliance constraints are embedded into the response process. This design not only reduces false alarm rates and shortens mean time to recovery (MTTR) but also promotes systematic knowledge accumulation and reuse within O&M processes.

The main contributions of this paper are summarized as follows:

(1) Enhanced anomaly detection and alert management system: We propose a Blended RAG-enhanced LLM multi-agent framework for power system O&M. By integrating Blended RAG into a multi-agent collaboration framework, the system implements a closed-loop workflow of detection - decision - validation. The framework is tailored and optimized for power O&M scenarios, significantly improving automation, factual consistency, and interpretability in anomaly handling.

(2) Safety compliance and validation mechanism: A dedicated safety verification agent is embedded into the workflow, which validates response plans across multiple dimensions including operation sequence, device state transitions, and personnel permissions. Combined with real-time SCADA interface monitoring, this mechanism effectively reduces operational safety risks and enhances system trustworthiness.

II. SYSTEM MODEL

This paper constructs a framework, as shown in Fig. 1, for a multi-agent collaborative business alarm disposal system, which is divided into six layers from the underlying data perception to the upper-layer business application. This architecture achieves the organic unity of data processing, anomaly detection, knowledge retrieval and decision execution through hierarchical functional division. While ensuring the maintainability of the system, it also has good adaptability to the power industry.

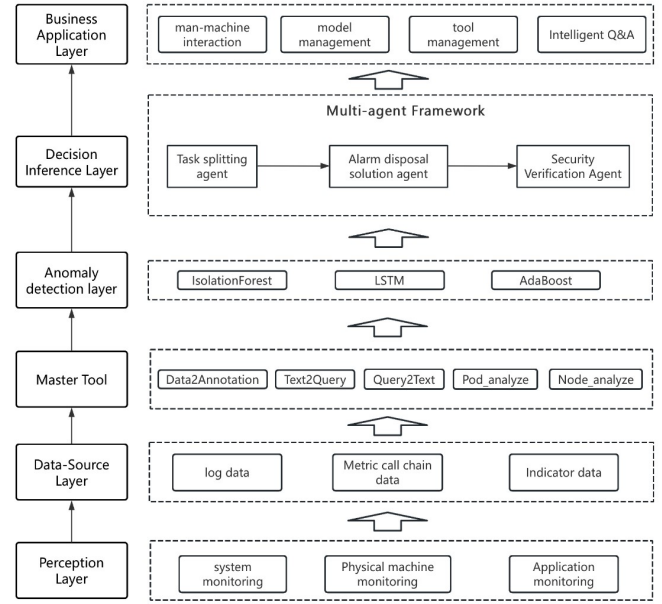


Fig. 1. Six layers architecture of the system

A. Perception Layer

As the entry point of the system, this layer is responsible for real-time monitoring of the operational state and multi-source data acquisition[10]. By integrating mainstream collection tools such as Filebeat, Telegraf, and OpenTelemetry, it provides comprehensive coverage of system logs, performance metrics, and business link data. For system logs, the collected content includes key infrastructure components such as middleware, databases, and container platforms, capturing exception stacks, service status changes, and error codes to support anomaly detection with rich contextual semantics. For performance metrics, the system gathers low-level resource data such as CPU utilization, memory usage, disk I/O, and thread counts in real time to monitor workload conditions and fluctuation trends. For business links, leveraging OpenTelemetry's distributed tracing capability, the system performs end-to-end tracking of service invocation chains, detecting response latency, jitter, and dependency failures, thus enabling proactive perception of potential fault propagation paths. All collected multi-source data are uniformly formatted and standardized, encapsulated via the OpenTelemetry Protocol (OTLP), and transmitted asynchronously through Kafka message queues. This ensures that the upper-layer model engine and knowledge processing modules receive stable, low-latency data supply, meeting the stringent real-time and accuracy requirements of power information systems in high-concurrency and high-availability environments[11].

B. Data-source Layer

The data layer establishes a multi-modal data storage system tailored to anomaly handling. It employs Elasticsearch to build a log index library for efficient field-based retrieval, constructs a vector knowledge base to store encoded semantic representations of work order documents, and uses InfluxDB for time-series storage and analysis of performance metrics. In

addition, a data annotation platform is deployed to provide structured samples for model training. A domain-adapted fine-tuning of Sentence-BERT based on the Grid Dispatching Terminology Standard significantly improves the recall of domain-specific terminology, laying a solid foundation for subsequent knowledge retrieval.

C. Tool Layer

The tool layer provides standardized data processing and transformation capabilities for the upper layers through five core tools that automate the preprocessing of multi-modal data:

Data2Annotation transforms raw logs into annotated structured data, using a combination of rule-based templates and deep learning to automatically identify entity boundaries and generate BIOES-compliant datasets. It supports domain-specific terminology mapping for power systems.

Text2Query converts natural language descriptions from operators into blended RAG-compatible queries, using a fine-tuned BERT model combined with a domain knowledge graph to ensure accurate translation of professional expressions.

Query2Text performs the reverse transformation, converting structured retrieval results into human-readable natural language descriptions through a hybrid template – neural generation approach.

All tools expose services via RESTful APIs and operate under an asynchronous pipeline to handle high-concurrency requests. Intermediate data are temporarily stored in Avro binary format within message queues and ultimately output as structured data streams for consumption by the anomaly detection layer, forming an efficient toolchain collaboration mechanism.

D. Anomaly Detection Layer

This layer performs in-depth analysis and intelligent scheduling of anomaly events. Using classification models, it identifies the type and severity of anomalies, categorizing them as hardware failures, network interruptions, application timeouts, etc., and labeling them with levels such as critical, severe, or warning. Based on a Neo4j graph database, the system constructs a service dependency topology to accurately assess the propagation impact of faults. Differentiated handling strategies are then triggered according to severity: critical anomalies immediately activate multi-agent collaborative response, while warning-level anomalies are stored in the knowledge base for later analysis. This layered processing ensures timely handling of major failures while avoiding unnecessary resource consumption.

E. Decision Inference Layer

This core layer implements intelligent and standardized anomaly handling, with all agent reasoning and action logic designed under the ReAct paradigm (Reasoning + Acting). Three core agents work collaboratively:

The task decomposition agent conducts causal analysis of anomalies, builds dependency graphs, and prioritizes actions based on power grid topology.

The response planning agent integrates retrieved knowledge to generate actionable, atomic response steps, each specifying the target object, execution instruction, and expected outcome.

The safety verification agent embeds a domain-specific safety knowledge base to validate compliance across multiple dimensions, including sequence of operations, device state transitions, and personnel authorization.

Dynamic interaction between reasoning and acting is achieved via carefully designed prompt templates, ensuring that LLM-generated reasoning chains conform to domain-specific logic and remain within predefined safety boundaries. Each decision cycle consists of four stages—anomaly analysis, knowledge retrieval, plan generation, and safety validation—with results recorded in a blockchain-based ledger for immutable audit trails. A feedback mechanism monitors recovery via SCADA APIs; if outcomes deviate from expectations, the system automatically triggers rollback and re-reasoning.

F. Application Layer

The application layer provides a human – AI collaborative O&M interface with transparent and auditable decision processes. Using a micro-frontend architecture, the system visualizes response workflows in timeline form, including root cause analysis, response rationale, and safety validation results. The work order system automatically generates standardized instructions, risk alerts, and fallback plans, which operators can confirm before execution. All operations are logged with metadata such as operator, execution time, and system state, ensuring that critical actions are traceable, reviewable, and auditable in compliance with power industry regulations.

III. BLENDED RAG-ENHANCED MECHANISM

A. Blended RAG-Enhanced Response Plan Generation Mechanism

Retriever-Augmented Generation (RAG) is a hybrid architecture that combines retrieval-based information access with language model generation, aiming to address the limitations of purely generative models in terms of knowledge coverage and factual accuracy. Its core idea is to retrieve semantically relevant content fragments from an external knowledge base before generating responses, and then provide these retrieved results as contextual input to the language model, thereby assisting it in producing more accurate and well-grounded answers. A conventional RAG framework consists of two modules: a Retriever and a Generator. The retriever typically employs dense embeddings (e.g., DPR or Sentence-BERT) to encode queries semantically and retrieve the most relevant knowledge documents in the vector space, while the generator adopts a Transformer-based large model that concatenates the query with retrieved content to generate the final response text.

In this paper, we propose a Blended RAG architecture that extends traditional RAG by integrating dual-channel retrieval, combining both sparse and dense methods, along with a Best Fields strategy and domain-specific corpus adaptation[10]. This design significantly improves recall and retrieval

precision for power-industry-specific terminology. Concretely, in the retriever module, the system deploys two subcomponents: (1) a sparse vector retriever based on the Elser model, enhanced with BM25-based inverted indexing for lexical-level term matching, and (2) a dense semantic encoder trained with Sentence-BERT, which captures implicit semantic relations between user queries and fault contexts. Within the blended structure, the sparse channel ensures precise recall of technical terms and template-based descriptions, while the dense channel captures extended natural-language problem descriptions and semantically ambiguous expressions. Their results are then merged using a vector re-ranking algorithm, augmented by business-weighted confidence scoring, ensuring that the most contextually relevant content fragments are prioritized as input to the generator.

Furthermore, to accommodate the highly structured nature of power information system data, the system applies the Best Fields strategy, which incorporates field-level weighting during vector encoding. This mechanism integrates structured business fields into the embedding process, aligning retrieval outputs more closely with specific operational contexts and avoiding irrelevant matches (e.g., superficially similar but unrelated interface errors).

For corpus construction, the system leverages historical work orders, equipment failure documents, safety operation guidelines, and monitoring alarm records collected from real-world power enterprises as the foundation of the knowledge base. The dense model is fine-tuned through domain adaptation, while the sparse indexing process is enhanced with a power-

industry terminology dictionary to strengthen term-frequency modeling[12]. As a result, the model gains sensitivity to domain-specific terms such as "soft switch failure" and "control center alert storm." Finally, in the generator module, prompt templates are used to standardize input formats, thereby improving both the interpretability and the actionability of generated results.

B. Layered Application of RAG in Multi-Agent Systems

To achieve intelligent alarm response and standardized incident handling in power information systems, this study designs a multi-agent collaborative architecture at the decision-making layer based on the ReAct paradigm (Reasoning + Acting), as shown in Fig. 2. Within this architecture, a RAG enhancement mechanism is systematically embedded. Unlike conventional approaches that treat RAG as an independent service component, our design integrates RAG capabilities deeply into three core agents according to task semantics. This enables comprehensive support across problem understanding, knowledge retrieval, action generation, and compliance validation, thereby significantly improving system response accuracy, knowledge utilization efficiency, and decision interpretability.

The task decomposition agent focuses on building anomaly causal chains and categorizing problem types. Upon receiving structured anomaly labels and log summaries from the perception layer, the agent invokes the RAG engine to construct query templates such as "event description + time window + host identifier." This retrieves historical fault

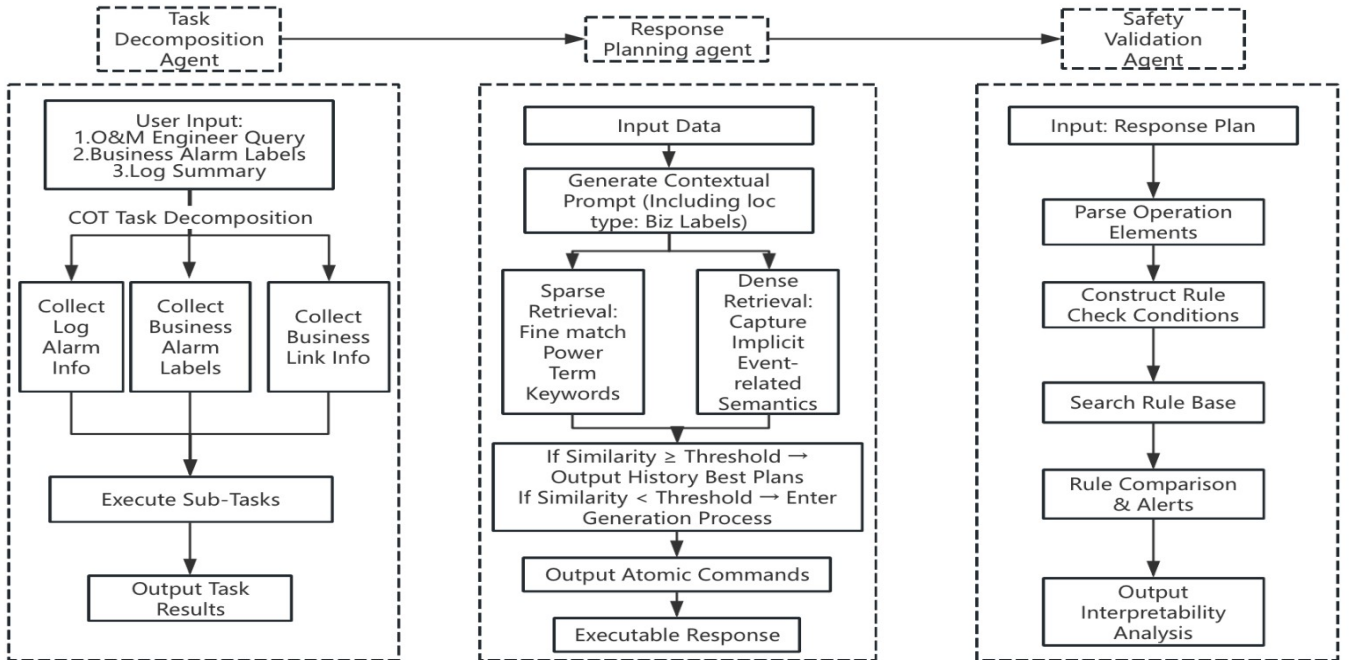


Fig. 2. Multi-agent collaborative architecture

samples and propagation records with similar characteristics. For example, in the case of a "SCADA master station heartbeat timeout" event, the system can automatically associate and supplement frequent upstream and downstream causal points to

form a preliminary event graph. In this process, RAG returns not only technical descriptions but also component call sequences and business impact analyses, thereby enabling the

transition from anomaly features to causal reasoning in the semantic space.

The response planning agent performs the end-to-end task of retrieving existing handling strategies and generating executable atomic instructions. Based on the results of task decomposition, the agent constructs contextual vector queries that activate both the sparse retriever and the dense retriever (for capturing natural-language semantics). Retrieved knowledge includes O&M manual fragments, expert work orders, and equipment documentation, which are then combined with fault context and fed into a BART generator. The output is a multi-step response plan containing operation objects, action types, and expected states. Each step provides interpretable semantics along with its knowledge source. In this process, RAG not only enhances the factual quality of plan generation but also constrains outputs through structured prompt templates, preventing the production of hallucinatory instructions.

The safety validation agent employs RAG to support rule retrieval, comparison, and dynamic policy enforcement. After receiving a response plan, the agent cross-checks each step against industry-standard operational guidelines to evaluate compliance, risk levels, and permission requirements. RAG provides both compliance judgments and explanatory analysis, ensuring that validation results are traceable and auditable.

In summary, the proposed RAG-enhanced mechanism achieves deep integration and functional embedding across the three agent dimensions of task decomposition, response planning, and safety validation. It leverages RAG's contextual retrieval capabilities while incorporating structural constraints, prompt templates, and semantic enhancement techniques, thereby ensuring that LLM reasoning processes remain professional, safe, and controllable. This mechanism constitutes the technical core for realizing the system's objective of "intelligent recognition - knowledge-supported reasoning - standardized execution."

IV. SIMULATION AND EXPERIMENTS

To validate the feasibility and effectiveness of the proposed Blended RAG-Enhanced LLM Multi-Agent System for Business Anomaly Detection and Alert Handling in power information system operation and maintenance (O&M) scenarios, this study designs three comparative schemes:

- (1) Baseline: Traditional dense-retrieval RAG (without multi-agent collaboration);
- (2) Multi-Agent + Traditional RAG;
- (3) Multi-Agent + Blended RAG.

All three schemes are deployed under the same distributed microservice architecture, ensuring that the upstream perception modules, knowledge base, and execution interfaces remain identical to guarantee fairness in comparison.

The experimental dataset is derived from real-world power dispatching systems, including historical alarm records, SCADA monitoring logs, and business telemetry data. Additional fault categories were introduced via fault injection, covering typical scenarios such as database connection failures,

control-center alert storms, SCADA heartbeat timeouts, network partitions, and container orchestration anomalies. In total, the dataset contains 90,000 labeled alarm events, of which 40,000 are used for domain-adaptive fine-tuning of dense and sparse retrieval models, while the remaining 50,000 are used for system performance evaluation. Each sample includes structured event metadata (event ID, affected component, timestamp, severity level), unstructured fault descriptions, and corresponding handling records, supporting the evaluation of both retrieval accuracy and response executability.

Evaluation metrics include precision and recall of alarm detection and handling, Mean Time To Recovery (MTTR) for measuring handling efficiency, and execution success rate (Exec-Rate), which reflects the validity of the response plans generated and executed by the system. The experimental results are summarized in TABLE I. and Fig. 3. The results indicate that while traditional RAG provides basic alarm handling capabilities, it shows clear limitations in recognizing domain-specific terminology and leveraging structured data in power O&M scenarios. This leads to lower retrieval precision, longer mean time to recovery (MTTR), and reduced execution success rates. With the introduction of a multi-agent framework, the system achieves significant improvements in precision, recall, and execution efficiency.

TABLE I. EXPERIMENTAL RESULTS

Method	Precision (%)	Recall (%)	MTTR (minutes)	Execution Success Rate (%)
Traditional RAG	86.2	82.7	21.4	78.5
Muti-agent+Traditional RAG	91.5	88.9	15.2	87.9
Muti-agent+Blended RAG	95.4	94.1	9.8	95.3

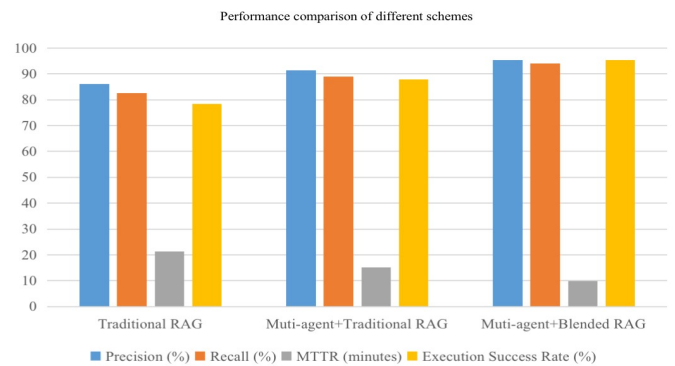


Fig. 3. Histogram of the Experimental result

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multi-agent framework, the system achieves significant improvements in precision, recall, and execution efficiency, owing to its capabilities in task decomposition, causal reasoning, and closed-loop decision-making. The proposed Multi-Agent + Blended RAG scheme performs best: it achieves precision and recall both above 94%, reduces MTTR by more than 50% compared to the baseline, and raises execution success rates to over 95%. These results validate its comprehensive advantages in ensuring factual consistency, enhancing executability of response plans, and improving O&M efficiency.

From the overall trend of experimental outcomes, the integration of a multi-agent architecture with Blended RAG outperforms traditional approaches across all performance metrics. Compared with single RAG methods, the multi-agent framework mitigates knowledge omissions and decision latency caused by single-point reasoning through task decomposition and collaborative execution, thereby improving both alarm recognition accuracy and handling efficiency. Building on this, the incorporation of Blended RAG further strengthens domain adaptability and domain-specific terminology recall during retrieval, ensuring that agents have access to higher-relevance and higher-confidence factual support during reasoning and generation. This not only alleviates the hallucination problem commonly observed in large models applied to specialized domains but also makes the generated response plans more robust in terms of executability and safety.

In essence, the fundamental reason for performance improvement lies in Blended RAG's ability to optimize both the breadth and precision of knowledge retrieval by combining sparse and dense strategies. This provides a stronger factual foundation for the reasoning chains of multi-agent systems, thereby forming a positive feedback loop of "high-quality retrieval – high-confidence reasoning – efficient execution."

V. CONCLUSION AND FUTURE WORK

In this paper, we address the challenges of inaccurate knowledge retrieval, fragmented decision chains, and limited executability of response plans in next-generation power information system operations and maintenance. We propose a Blended RAG-Enhanced LLM Multi-Agent System for Business Anomaly Detection and Alert Handling. By introducing a fusion mechanism of sparse and dense retrieval into the multi-agent collaboration framework, the system significantly improves domain knowledge recall and factual consistency. Together with agent modules for task decomposition, response planning, and safety validation, it realizes a closed-loop workflow of "detection–decision–verification."

Experimental results demonstrate that the proposed approach outperforms traditional solutions in terms of precision, recall, alarm-handling efficiency, mean time to recovery (MTTR), and execution success rate. These results

validate the effectiveness and feasibility of the system in enhancing automation, reducing hallucination risks.

Future work will be expanded in two directions. First, cross-modal information fusion will be introduced into the knowledge retrieval stage, integrating images, topological structures, and log texts to enable multi-source information-driven knowledge enhancement. Second, we will explore the combination of large language models with online learning mechanisms, enabling the system to dynamically update its knowledge base and reasoning strategies during long-term operation, thereby addressing emerging fault patterns and sophisticated attack methods in power information systems. Through these improvements, we aim to further enhance the adaptability, robustness, and practical value of the system in real-world power O&M applications.

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